

Ali Salehi Darjani

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Education

Ph.D. in Electrical and Computer Engineering	2025 – Present
<i>University at Albany, SUNY</i>	<i>GPA: 4.00 / 4.00</i>
B.Sc. in Electrical Engineering	2018 – 2023
<i>K. N. Toosi University of Technology, Tehran, Iran</i>	<i>GPA: 3.33 / 4.00 (16.51 / 20) Last 2 Years GPA: 3.80 / 4.00</i>

Technical Skills

Programming Languages:

Python (advanced), C++, C

Machine Learning & Deep Learning:

PyTorch, TensorFlow, scikit-learn, NumPy, computer vision pipelines, supervised and deep neural networks

Data & Scientific Computing:

Pandas, Matplotlib, OpenCV

Systems & Development Tools:

Linux (Ubuntu), Bash, Git, GitHub, Slurm, tmux

Scientific Writing & Documentation:

L^AT_EX

Engineering & Productivity Tools:

MATLAB, Microsoft Office (Word, PowerPoint, Excel)

Embedded Systems & Hardware:

Raspberry Pi, microcontrollers (STM32, Arduino)

Electronic Design & Simulation:

Keysight ADS, LTspice, CST Studio Suite

Research Interests

- Machine learning
- Statistical and probabilistic learning
- Sequential decision-making (MDPs, POMDPs)
- Decision-making under uncertainty
- Mathematical foundations and applications to large language models and generative models
- Reinforcement learning

Research Experience & Projects

POMDP-Based Dynamic Cascades of Deep Learning Models

2025 – Present

Developing POMDP-based frameworks for sequential decision-making under uncertainty, focusing on dynamic selection and cascading of deep learning models for cost-efficient inference.

Supervisors: Dr. Daphney-Stavroula Zois

Cascades of Heterogeneous Models for Efficient Credit Assessment

2025

Designed instance-wise cascades of heterogeneous machine learning models for cost-aware inference in credit assessment, with emphasis on learning from refined noisy data and accuracy–cost trade-offs.

Supervisors: Dr. Daphney-Stavroula Zois, Dr. Charalampos Chelmis

Children ADHD Detection Using Multimodal Data

2023

Conducted research on automated early detection of Attention-Deficit/Hyperactivity Disorder (ADHD) in children using machine learning and deep learning techniques.

Supervisor: Dr. Farnaz Sheikhi

Bachelor's Thesis: Vision-Based Sleep Detection on Embedded Systems

2023

Designed a computer vision-based sleep detection algorithm using Python and image processing techniques, and deployed the system on a Raspberry Pi platform for real-time inference.

Supervisor: Dr. Mahmoud Ahmadian Attari

Publications

- **Learning to Cascade: A POMDP Approach to Sequential Model Selection**
Ali Salehi Darjani, Daphney-Stavroula Zois
ICASSP 2026 (Accepted)
- **CHEX: A Cascade of Heterogeneous Experts for Instance-Wise Credit Assessment**
Zois, Chelmis, Azarshab, Ali Salehi Darjani
IEEE International Conference on Big Data (BigData), 2025 (Published)
DOI: 10.1109/BigData66926.2025.11402635

Teaching Experience

Teaching Assistant, Digital Signal Processing & Electronics	(Jan 2026 – May 2026)
Teaching Assistant, Engineering Statistics & Electronics	(Sep 2025 – Dec 2025)
Teaching Assistant, Machine learning & Electrical Circuits	(Jan 2025 – May 2025)

References

Dr. Daphney-Stavroula Zois, Professor

Department of Electrical and Computer Engineering, University at Albany, SUNY

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